

Vedant Singh Rathore

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Education

- **University College London** 09/2024 - 09/2025
MSc: Machine Learning
Thesis: Probabilistic Regression for Jet Energy Calibration, ATLAS, CERN
- **St. Stephen's College, Delhi University** 07/2017 - 08/2020
B.Sc. (Hons.) Physics with a minor in CS
GPA: 8.76/10

Scholar Honours

- **Chevening Scholar, 2024-25**: Chosen for the prestigious Chevening Scholarship for 2024-25 cohort by UK Government's Foreign, Commonwealth and Development Office. Rewarded with a 100% scholarship to pursue MSc.
- **INSPIRE Scholar, 2017-2020**: Chosen for the INSPIRE Scholarship, Department of Science and Technology, Government of India. Covered 100% Tuition Fees for undergraduate education.

Research Projects

- **Memory Augmented Self-Improvement for STLMs (PolyMind)** — *UCL, Stat NLP*:
Designed a **multi-agent fine-tuning framework** for compact LMs (77M) using **LoRA adapters** and a **persistent memory buffer** for critique storage. A separate **4B judge model (Gemma)** provided feedback, later converted into instruction-tuning prompts. Demonstrated **statistically significant performance gains** on the SAMSum summarization dataset (**ROUGE-1 F1: 0.393 → 0.406**) via memory-based refinement without human supervision.
- **Retrieval-Augmented NetHack Agent (wiki-BALROG)** — *UCL, Open-Endedness and General Intelligence*:
Enhanced NetHack agents by integrating **Retrieval-Augmented Generation (RAG)** into the BALROG benchmark using **FAISS indexing** and **AutoRAG query generation** from LLM observations. Summarized retrieved context before passing it to the agent, enabling more focused reasoning and improved decision-making. Achieved a **progression score of 1.57**, surpassing prior leaderboard entries.
- **Weakly-Supervised Semantic Segmentation** — *UCL, Applied Deep Learning*:
Trained U-Net and ResNet50 using CAM-based pseudo-labels with weak supervision (image labels, boxes, scribbles). Achieved 87.5% mean IoU (95% of full supervision) with 95% fewer annotations.

Experience

Marico Limited - Mumbai, India 09/2020 - 08/2024

Data Science Manager

Managed a team of 3 Data Scientists and 1 Analyst where the KRAs included but are not limited to:

- **Audio Analytics Tool**: Created and managed in-house audio analytics tool for transcription, translation, sentiment analysis, conversation summary and chatbot. Leveraged Azure STT models and OpenAI models.
- **Forecast Engine**: Developed an in-house product for automated end-to-end Forecasting comprising of EDA, Transformations, Modelling & Output Preparation. Reduced total runtime of Demand Forecasting by 70%.
- **Demand Forecasting**: Create and manage monthly demand forecasts and production plans for 3 global markets providing necessary inputs to the planning & distribution team. Involves CXO level stakeholder management. Accuracy improved by 12% for India geography. Designed dashboards to provide actionable insights into business performance.
- **Range Based Forecasting**: Created a mathematical model to predict volume ranges with a given confidence interval using residual bootstrapping. Led to 18% bias reduction to prevent over-stocking.
- **ONDC**: Developed strategy & led the team to onboard Marico onto the Open Network for Digital Commerce, establishing a new sales channel with increased customer reach. Onboarding all brands onto the network to enable digital commerce and collect 1P data.
- **OpenAI**: Created inhouse chatbots, summarization tool, code helper and more using Azure OpenAI APIs. Created Chatbot solution where employees can upload internal data and get outputs. Tools being used for CXOs summaries as well.
- **DTC Management**: Led a DTC business to generate 1P user data and optimize business for bottom-line growth. Improved bottom-line as percent of top-line from -65% to +13%. 1P data used to derive customer behavior insights, make cohorts, and create strategies for brand marketing.

Skills

Python, Git, Github, Machine Learning, Regression, Hypothesis Testing, Statistical Data Analysis, Predictive Modelling, SQL, Supply Chain Analytics, Code Review, Analysis and Modeling, Process Modeling, Agile Methodologies, Time-Series Analysis

Certifications

- **Quantum Algorithms for Cybersecurity, Chemistry, and Optimization**: MIT xPro (05/2024 - 06/2024)
- **Quantum Computing Fundamentals**: MIT xPro (04/2024 - 05/2024)
- **Data Science & Machine Learning**: MIT Schwarzman College of Computing (03/2023 - 07/2023)